SPE-226875-MS

Fiber Optics Reservoir Monitoring System Sustainability in Extreme Gas Well Conditions

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

The purpose of this study is to illustrate how the long-term viability of fiber optic technology in conjunction with a Permanent Downhole Monitoring System can considerably endure the difficult conditions that are present in wells that are subjected to severe vibration and extreme pressure. The system ensures that defined target objectives are consistently delivered, which adds value to a large middle eastern National Oil Company (NOC). This is accomplished within the challenging conditions of gas production wells.

One of the most persistent challenges that arises during routine operations is the deployment of a downhole pressure and temperature gauge that is both dependable and sustainable in wells that are subject to high levels of pressure and vibration. Because of the intense vibrations and pressure that are exerted on the electronic components of a normal electronic downhole monitoring gauges, electronic systems that are deployed downhole are frequently prone to failure. With its ability to perform well under high-pressure, high-temperature, and high-vibration circumstances, the optical pressure and temperature (P&T) gauge system, in conjunction with Distributed Fiber Optic Sensing (DFOS), provides a solution that is reliable.

Optical P&T systems have been thoroughly integrated into a variety of field applications, and additional systems are planned to be deployed both onshore and offshore in the near future. After they have been installed, these technologies will transfer vital information in real time to the control room of the engineering team as well as to the teams responsible for reservoir management. Using fully developed fiber optic systems, which alleviate prevalent operational difficulties such as high vibration settings, harsh pressures, and extreme temperatures condition survivability, it is possible to guarantee reliable and superior data transmission even in challenging downhole conditions. This is made possible through the utilization of fiber optic systems. Through real-time data access and advanced analytics, which include well integrity acknowledgments, the inclusion of a monitoring system that is enabled with DFOS considerably improves the supply of crucial insights into the integrity of the well (Gang Tao, 2023). It is anticipated that the combination of DFOS and downhole monitoring gauges will result in long-term benefits. These benefits will be achieved through the enhancement of performance, the prediction of possible failures, and eventually the maximization of the well system's longevity and efficiency.

The integration of DFOS technology with downhole monitoring gauges results in the creation of a cutting-edge monitoring solution that generates data in real time on the dynamics of reservoirs under extreme conditions, such as those that are encountered during fracturing applications. The system is positioned as a new solution for proactive reservoir management as a result of these qualities. It provides novel chances for optimization and enhances operational decision-making to an extent that has never been seen before.

Keywords: Fiber Optics, Permanent Downhole Monitoring System, Gas Wells, Completions, Reservoir Monitoring

Introduction

The oil and gas sector is always looking for innovative monitoring solutions in order to improve production efficiency, guarantee the integrity of wells, and maximise reservoir management. Conventional electronic-based Permanent Downhole Monitoring Systems frequently experience dependability concerns when operating in harsh gas well conditions. These conditions are characterised by high pressure, high temperature (HPHT), and intense vibrations. A durable alternative has evolved in the form of fiber optic sensing technology, which provides improved durability and real-time data collecting in extreme settings such as those found in downhole environments.

In this paper, we explore the possibility of using a fiber optic-based downhole monitoring gauges in conjunction with Distributed Fiber Optic Sensing (DFOS) in order to circumvent the restrictions that are associated with conventional electronic gauges. The study focusses on the case implementations, operational benefits, and long-term sustainability in gas wells that are operated by a major NOC in the Middle East.

Challenges in Conventional Downhole Monitoring Systems

Limitations of Electronic Downhole Monitoring Gauges

When implemented in harsh gas well settings, conventional downhole monitoring systems, particularly those based on electronic Permanent Downhole Monitoring Systems, face considerable constraints. This is especially true for those systems. These systems are highly dependent on delicate electronic components that are susceptible to malfunctioning when subjected to situations that involve high vibration, high pressure, and high temperature (HPHT) conditions (S. Busetti, V. Kazei, H. Merry, 2024). With having minimum hundreds of components downhole, the sensor deterioration and electronic drift are symptoms that frequently occur as a result of the extreme mechanical stress that is caused by gas flow and artificial lift systems. In addition, electromagnetic interference (EMI) in the surroundings of the downhole might affect the integrity of the signal, which can lead to erroneous data transfer. Because of these vulnerabilities, the system has frequent failures, which in turn necessitates expensive workovers and increases the amount of time that is not productive (NPT), which ultimately leads to an increase in operational expenses. Each component, from microprocessors to pressure transducers and signal amplifiers, must operate flawlessly under extreme conditions. However, thermal cycling, mechanical fatigue, and chemical exposure can degrade these components over time, leading to cascading failures. The diagnostic process for such failures is often time-consuming and requires specialized equipment and personnel, further increasing downtime. Moreover, the need for periodic recalibration and maintenance of electronic systems adds to operational complexity. These limitations underscore the necessity for more resilient alternatives, such as fiber optic-based systems, which offer a simplified architecture with fewer failure-prone elements and greater tolerance to harsh downhole environments.

Need for a Robust Alternative

In order to address these difficulties, there is an urgent need for an alternative that is more substantial and dependable, and that is capable of withstanding the extreme conditions that are present in gas wells. For a monitoring system to be considered optimal, it must be able to provide continuous real-time data on pressure and temperature, as well as demonstrate resistance to both mechanical and thermal challenges. A further need is that it should be able to survive for an extended period of time without requiring frequent maintenance or intervention. The system will be having highest risk of failure during installation as shown in Fig. 1. The technology of fiber optics has emerged as a potentially useful answer since it provides inherent immunity to electromagnetic interference (EMI), remarkable endurance, and the capacity to perform dependably in harsh conditions. Considering these characteristics, fiber optics is an excellent option for downhole monitoring in environments that provide a number of challenges.

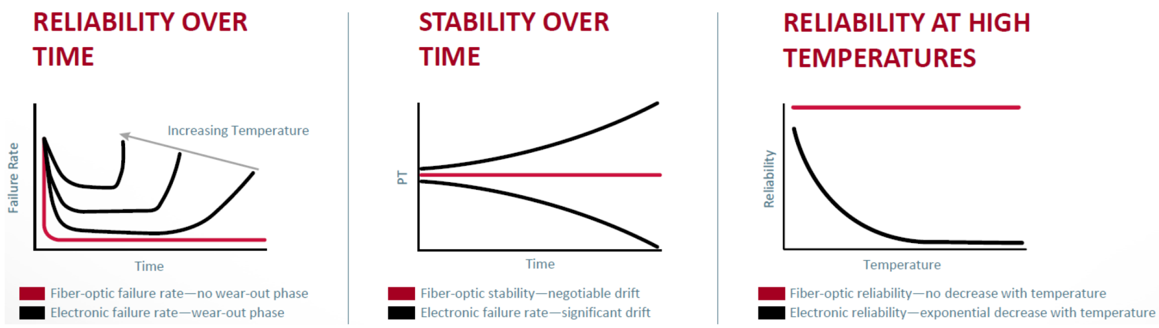

Figure 1—Reliability of In-Well Fiber Optics System Deployment

Fiber Optic Monitoring System: Technology Overview

Optical Pressure and Temperature (P&T) Gauges

Particularly through the utilization of optical pressure and temperature (P&T) gauges, fiber optic monitoring systems as in Fig. 2 which are constructed with single-mode and multimode fibers as shown in Fig. 3 represents a substantial breakthrough in the technology that is used for downhole sensing. Bragg grating technology is the basis for the operation of these gauges. This technology is based on the idea that changes in pressure and temperature induce adjustments in the wavelength of light that is reflected. Utilizing this optical method eliminates the requirement for downhole electronics, which in turn eliminates the primary source of failure that is present in conventional systems. Because it does not contain any electrical components, the system is more durable, particularly in situations that are subject to high levels of vibration. In addition, fiber optic gauges provide higher precision and long-term stability, which makes them an excellent choice for protracted deployments in wells that can achieve temperatures of up to 300 degrees Celsius.

Figure 2—Fiber Optic Gauge System

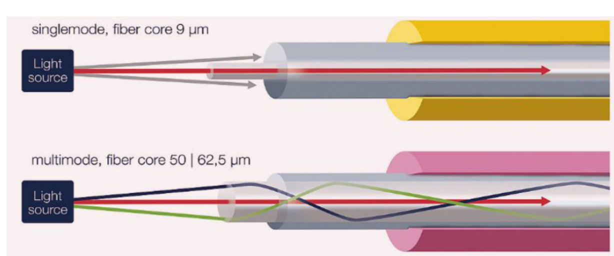

Figure 3—In-Well Fiber Optics Cable Construction

Distributed Fiber Optic Sensing (DFOS)

Distributed fiber optic sensing, also known as DFOS, is a technique that enhances the capabilities of fiber optic systems. It does this by permitting continuous monitoring over the full length of the wellbore, in addition to point sensing. The distributed temperature sensing (DTS) and Distributed Acoustic Sensing (DAS) technologies are included in the distributed field-based observation system (DFOS). It is crucial to have a comprehensive understanding of thermal dynamics and fluid flow inside the reservoir, and DTS gives accurate temperature profiles regarding these aspects. Analyzing the acoustic signals that are captured by the DAS, on the other hand, allows for the detection of fluid flow, the identification of leaks, and the monitoring of sand output. These capabilities of distributed sensing provide a full view of the circumstances of the well, which improves both our knowledge of the situation and our ability to make decisions on operations.

System Integration with Downhole Monitoring Gauges

DFOS and downhole monitoring gauges working together to create a synergistic monitoring solution that offers increased data reliability even when subjected to hard conditions is the consequence of the combining of these two monitoring systems. This combo system as utilizing the Raman and Rayleigh Scattering as shown as build up cable contruction in Fig. 4 offers assistance for predictive maintenance by continuously monitoring well parameters and identifying anomalies before they get more serious and lead to failures. This can be accomplished through the use of this system. Real-time analytics, which are made feasible by this integration, also make it possible to improve reservoir management. This is a significant benefit. The capacity to make well-informed decisions that maximize production and extend the well's lifespan is provided to operators as a result of this.

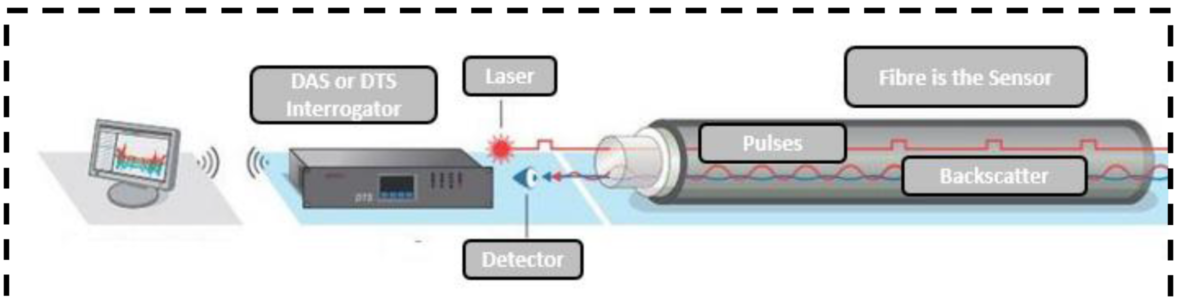

Figure 4—Distributed Fiber Optic Sensing (DFOS) System for DTS and DAS

Field Applications and Case Studies with Gas Well Deployment

The usefulness and dependability of fiber optic downhole monitoring gauges in real-world settings have been proved in a number of field deployments, which provide evidence of its practical applicability. These deployments have been undertaken in a variety of scenarios. The use of the technology has provided evidence that this is the case. Fiber optic systems were put in gas wells that were typified by high levels of vibration. These deployment was one of the most significant deployments ever carried out. The system revealed that there were no failures during the course of a monitoring period that lasted for twenty-four months, which is a striking contrast to the thirty percent failure rate that is typically reported with traditional electronic gauges. Certainly, this was an impressive accomplishment. In addition, the deployment made it possible to monitor the processes of hydraulic fracturing in real time, which resulted in the collection of significant data that was utilised to guide and optimise the designs of completions. The survivalibility of the system exceed expectations to number of years which is significant to cost implication. Utilising this data, completion designs were guided and optimised for optimal performance. By collecting the dynamic reservoir responses that occurred during stimulation, engineers were able to refine their approaches and further boost the overall performance of the well. This was accomplished through the process of collecting the responses.

Sustainability and Long-Term Benefits

The adoption of fiber optic downhole monitoring gauges contributes significantly to the sustainability and efficiency of well operations. One of the primary benefits is the reduction in intervention costs, as the extended lifespan of fiber optic gauges minimizes the need for frequent replacements or repairs. This durability translates into fewer workovers and less downtime, which not only lowers operational expenses but also enhances overall productivity. The system's ability to support predictive failure analysis further contributes to operational efficiency by enabling proactive maintenance strategies that prevent unexpected disruptions. From a reservoir management perspective, the continuous data provided by DTS and DAS facilitates dynamic characterization of reservoir behavior. This real-time insight allows for more accurate well placement and the development of optimized production strategies tailored to evolving reservoir conditions. The granular data captured by fiber optic sensors supports a deeper understanding of fluid flow patterns, pressure changes, and thermal dynamics, all of which are critical for maximizing hydrocarbon recovery.

In addition to operational and reservoir benefits, fiber optic monitoring systems offer notable environmental and safety advantages. The reduced frequency of well interventions lowers the exposure of personnel to hazardous conditions, thereby improving health, safety, and environmental (HSE) outcomes. Moreover, the system's capability to detect leaks at an early stage enhances environmental protection by preventing the release of hydrocarbons into surrounding ecosystems. These attributes align with the industry's growing emphasis on sustainable and responsible resource development. Furthermore, the integration of fiber optic downhole monitoring gauges with advanced analytics platforms enables operators to implement condition-based maintenance strategies. By continuously monitoring key parameters and identifying deviations from normal operating conditions, the system allows for timely interventions that mitigate the risk of equipment failure. This predictive approach not only extends the lifespan of critical assets but also ensures uninterrupted production, thereby improving the overall economic performance of the well.

The long-term sustainability of fiber optic systems is also supported by their minimal energy requirements and low environmental footprint. Unlike electronic systems that may require power-intensive components and frequent servicing, fiber optic sensors operate passively and require minimal maintenance. This makes them particularly suitable for remote or offshore installations where access is limited and operational efficiency is paramount.

Moreover, the high-resolution data generated by fiber optic systems can be integrated with digital twin models and reservoir simulation tools to enhance predictive capabilities. This integration allows for more accurate forecasting of reservoir behavior, improved planning of field development strategies, and better risk management. As the industry continues to embrace digital transformation, the role of fiber optic monitoring systems in enabling data-driven decision-making will become increasingly vital. The sustainability and long-term benefits of fiber optic downhole monitoring gauges in gas wells are multifaceted, encompassing operational efficiency, predictive maintenance, environmental stewardship, and advanced reservoir management. These systems not only address the limitations of traditional monitoring technologies but also pave the way for a more resilient and intelligent approach to well operations in extreme environments such as in gas production wells.

Novel Contributions and Industry Impact

A revolutionary step forward in the field of reservoir monitoring is represented by the utilization of Distributed Fiber Optic Sensing (DFOS) in conjunction with Permanent Downhole Monitoring Systems.

This cutting-edge combination makes it possible to perform real-time monitoring in gas wells that is vibration-tolerant for the very first time (S. Busetti, V. Kazei, H. Merry, 2024). This solution addresses a problem that has been plaguing the industry for a very long time. By taking advantage of the one-of-a-kind characteristics of fiber optics, the system is able to provide uninterrupted data with a high level of quality even in the most demanding operational situations.

One of the most revolutionary features of this technology is its ability to facilitate proactive reservoir management by utilizing artificial intelligence (AI) and advanced analytics. This is one of the most groundbreaking components of this technology. It is possible to use artificial intelligence algorithms to interpret the real-time data that is created by DFOS in order to recognize trends, anticipate breakdowns of equipment, and optimize production parameters (Gonzalez Gomez, 2025). With this capability, operators are given the option to make decisions based on data, which has the potential to improve overall asset performance, minimize risks, and enhance efficiency.

The traditional activities of oil and gas exploration and production are not the only ones that could potentially benefit from the utilisation of fiber optic downhole monitoring gauges in the future applications. When it comes to developing industries, such as geothermal energy and carbon capture, utilisation, and storage (CCUS), respectively, where dependable monitoring under difficult conditions is of equal importance, the method is ideally suited for use (Jonathan Ajo-Franklin et al, 2025). It is anticipated that the use of fiber optic sensing systems will play a significant part in facilitating the development of resources in a manner that is safer, more effective, and more environmentally friendly across a wide range of applications (Christopher Baldwin, 2018). This is due to the fact that the energy business is continuing to go through expansion, and it is anticipated that their adoption will play a significant role in this respect.

Conclusion

Fiber optic-based downhole monitoring gauges with DFOS represents a transformative advancement in downhole monitoring, particularly in extreme gas well conditions. By eliminating the weaknesses of electronic systems, this technology ensures long-term reliability, enhances reservoir insights, and reduces operational costs. As the industry moves toward smarter and more sustainable solutions, fiber optic monitoring systems are poised to become the standard for high-risk well applications. The transformative impact of this technology lies in its ability to deliver continuous, high-resolution data in real time, even under the most extreme pressure, temperature, and vibration conditions. This capability not only improves operational decision-making but also supports predictive analytics and proactive maintenance strategies, which are essential for minimizing downtime and maximizing asset performance. Industry adoption trends indicate a growing preference for fiber optic systems, particularly in regions with mature fields and complex well architectures. Major oil and gas operators are increasingly integrating DFOS-enabled downhole monitoring gauges into their digital oilfield strategies, leveraging the technology's compatibility with AI-driven analytics and cloud-based monitoring platforms. This integration facilitates seamless data acquisition, interpretation, and visualization, enabling operators to optimize production and enhance reservoir management in real time. Furthermore, the long-term value of fiber optic monitoring systems extends beyond operational efficiency. Their ability to reduce environmental risks, improve health and safety outcomes, and support regulatory compliance makes them a critical component of responsible resource development. As the energy sector continues to evolve toward decarbonization and digital transformation, fiber optic-based downhole monitoring gauges with DFOS will play a pivotal role in shaping the future of well monitoring and reservoir optimization.

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